Dual-Model Ensemble with Adaptive Normalization for Neural TSP Solvers

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Abstract

We present a dual-model ensemble approach for neural Traveling Salesman Problem (TSP) solvers that combines two independently trained Policy Optimization with Multiple Optima (POMO) models with complementary strengths. Model A is refined via preference-based post-training with curriculum learning, with its inference strategy optimized by LLM4AD-driven portfolio search. Model B is trained through an off-policy pipeline with LLM4AD-guided data engineering and geometry-aware data synthesis. Our adaptive InstanceNorm calibration mechanism dynamically assigns augmentation viewpoints between the two models under a fixed 8-viewpoint budget. On a held-out test set of 10 TSPLIB instances, the ensemble achieves an average optimality gap of 0.478%, a 76.7% relative improvement over the original POMO baseline. Ablation studies demonstrate that each component contributes meaningfully and that tour-level ensemble of diverse policies is essential for sequence decisions.

Keywords: Traveling Salesman Problem, Neural Combinatorial Optimization, POMO, Preference Learning, Model Ensemble, Test-Time Adaptation

1 Introduction

The Traveling Salesman Problem (TSP) is a classic NP-hard combinatorial optimization problem. Neural combinatorial optimization (NCO) has emerged as a promising paradigm that leverages deep learning to learn heuristic policies generalizing across problem instances. Among NCO methods, Policy Optimization with Multiple Optima (POMO)[1] achieves strong results by exploiting permutation invariance: starting from every city simultaneously and selecting the best tour among N parallel trajectories at zero additional inference cost.

Despite its strong performance, the original POMO exhibits limitations on real-world TSPLIB benchmarks. Its generalization to large-scale instances (100+ nodes) and non-uniform geometric structures remains suboptimal, with an average optimality gap of 2.048% on our test set. The standard training pipeline relies on uniform random data generation, which fails to cover the diverse geometric patterns encountered in practical TSP instances.

We address these limitations through a dual-model ensemble framework. Model A is refined from a pretrained POMO checkpoint via preference-based post-training with curriculum learning, with its inference strategy optimized by LLM4AD-driven portfolio search. Model B is trained from scratch through an off-policy pipeline that leverages LLM4AD-guided data engineering and LKH3 expert trajectories. We then propose an adaptive InstanceNorm calibration mechanism that dynamically assigns augmentation viewpoints between the two models under a fixed 8-viewpoint budget per instance. On a held-out test set, the ensemble achieves an average optimality gap of 0.478%, a 76.7% relative improvement over the original POMO baseline.

2 Related Work

Neural Combinatorial Optimization. Attention-based encoder-decoder architectures[2] have become the dominant paradigm for NCO. POMO improves upon this by using multiple starting points and a shared encoder, achieving strong zero-shot generalization. Subsequent works have explored heatmap-based approaches[3] and hybrid methods combining neural policies with local search[4].

Preference Learning. Direct Preference Optimization (DPO)[5] enables fine-tuning from preference pairs without explicit reward modeling. We adapt this framework to combinatorial optimization, where preferences are derived from tour length comparisons. Curriculum learning[6] progressively increases problem difficulty during training, preventing catastrophic forgetting of base capabilities.

Test-Time Adaptation. Techniques such as Test-Time Normalization (TTN)[7] recalibrate normalization statistics at inference to mitigate domain shift. We extend this idea to InstanceNorm layers in the transformer encoder, using Wasserstein distance to measure feature distribution shift and adaptively blending source and target statistics.

Large Language Model for Algorithm Design (LLM4AD). LLM4AD[8] provides a unified platform for automatic algorithm design with large language models. In this work, we leverage LLM4AD not to generate new algorithms, but to search inference strategies and evaluate data engineering proposals before committing GPU resources.

3 Methodology

Our approach consists of two independently trained models and an adaptive ensemble mechanism. Model A (6 layers, 128 embedding dim, 1.27M parameters) excels on medium-scale instances (101–150 nodes). Model B (8 layers, 192 embedding dim, 3.74M parameters) dominates large-scale instances (>200 nodes). Both models are based on the standard POMO transformer encoder-decoder architecture with InstanceNorm and 8-fold coordinate augmentation.

3.1 Model A: Preference Post-Training with Curriculum Learning

Starting from a pretrained POMO checkpoint on 100-node instances, we create two copies: a current policy π_θ and a frozen reference policy π_ref. The reference remains fixed throughout training, serving as a stable preference baseline against which the current policy is optimized.

Preference Pair Construction. For each TSP instance x, we collect candidate tours from both the current policy and the reference policy. These are ranked by reward $r(x,y) = -d(x,y)$, where d denotes the tour length. We form preference pairs by selecting the top-k tours as positives (y+) and bottom-k as negatives (y−).

$$Y(x) = Y_{policy}(x)$$
$$\cup\, Y_{ref}(x), y^{+\in Top-k(Y(x))}, y^{-\in Bottom-k(Y(x))}$$

DPO-Style Preference Loss. The preference loss is formulated as:

$$\delta = \beta$$
$$L_{pref} = -log\ \sigma(\delta), L_{total} = L_{pref} + 0.2 \times L_{RL}$$

where β = 0.1 is the temperature coefficient and L_RL is a standard REINFORCE loss with weight 0.2 for optimization stability. Key hyperparameters: k = 4 preference pairs per instance, preference loss weight = 1.0, and RL stabilization weight = 0.2.

Curriculum Learning. Training spans 60 epochs divided into three stages. Stage 1 (epochs 1–20) focuses on 150-node instances with 12.5% 100-node replay. Stage 2 (epochs 21–40) advances to 200-node with 20% 150-node and 10% 100-node replay. Stage 3 (epochs 41–60) targets 300-node with 20% 200-node and 10% 100-node replay. This progressive scaling prevents catastrophic forgetting of base capabilities while building large-scale generalization.

3.2 Model B: Off-Policy Training with Geometry-Aware Data

Model B follows an off-policy training pipeline that combines behavior cloning from expert trajectories with reinforcement learning fine-tuning on targeted geometric distributions.

LLM4AD-Guided Data Engineering. We employ an LLM4AD-derived sub-agent to systematically evaluate candidate data engineering strategies before committing GPU time. The sub-agent scores proposals based on geometric shape coverage (elongated, boundary-heavy, cluster-sparse, noisy-grid), size distribution, and risk controls (no TSPLIB leakage, no unsafe optimizer resumes). The highest-scoring strategy directs an iterative recipe tuning process: the pilot recipe (30% elongated, 25% boundary-heavy, 25% cluster-sparse, 10% uniform, 10% noisy-grid) is refined to v2 after intermediate checkpoint evaluation, adding n50/n75 protection and reducing elongated pressure to 15% to preserve nearby-domain capability.

Training Stages. The pipeline consists of three phases: (1) Behavior Cloning (BC) pre-training for 100 epochs on LKH3-generated expert trajectories across clustered, holey, boundary, and uniform distributions; (2) On-policy RL fine-tuning for 360 epochs with encoder freeze warmup and a 5-phase curriculum; (3) Geometry-aware fine-tuning from a pilot checkpoint with v2 data distribution: 25% random, 20% grid, 20% clustered, 20% boundary, 15% elongated, plus n50/n75 small-scale protection and maximum scale of 280 nodes. The learning rate is 5e-6 for 20 epochs with checkpoints saved every 5 epochs. Epoch 15 is selected as the optimal point based on intermediate evaluation.

3.3 Adaptive Dual-Model Ensemble

Beyond model training, we further optimize the inference procedure through an LLM4AD-guided strategy search framework. The goal is to identify effective test-time decoding portfolios for fixed checkpoints without modifying model weights. This makes the proposed enhancement purely inference-stage and directly compatible with pretrained POMO-style solvers.

The search space consists of plain argmax decoding and norm-calibrated forward passes applied to 18 candidate normalization groups, including individual encoder layers from layer0 to layer7, early and late layer subsets, and norm1/norm2 variants. For each candidate strategy, calibration statistics are collected at inference time and used to recalibrate the corresponding InstanceNorm modules. The resulting tour candidates are evaluated on a validation set using avg_aug_gap as the ranking metric.

For Model A, the LLM4AD-guided portfolio search selects a compact strategy that combines plain argmax decoding with layer4 normalization calibration. This strategy consistently improves over plain-only decoding while keeping the inference procedure simple and lightweight.

For Model B, the same search framework applied to seek better performance. A greedy portfolio construction algorithm iteratively adds the best-performing norm-calibrated candidate until convergence. The final compact5 strategy consists of plain argmax decoding plus four norm-calibrated forward passes using layer4, late_norm2, layer1, and early_norm2. During inference, the shortest tour among all candidates is selected as the final solution.

Overall, the LLM4AD-guided inference search provides a systematic way to discover effective test-time portfolios under fixed checkpoints. It complements training-stage improvements by exploiting normalization-sensitive behaviors of the learned policies, while avoiding any additional model training or weight modification.

3.4 Adaptive Dual-Model Ensemble

Complementarity Analysis. Model A and Model B are independently trained by different authors, exhibiting complementary strengths across problem scales. Model A (6 layers, 128 dim) excels on medium-scale instances (101–150 nodes), winning 7 out of 8 cases. Model B (8 layers, 192 dim) dominates large-scale instances (>200 nodes), winning all 3 cases with an average gap advantage of −3.17%. This complementarity motivates a tour-level ensemble rather than step-level fusion, where each model independently generates complete tours and we select the shortest among all candidates.

Step-Level Fusion Failure. We empirically verify that step-level fusion strategies catastrophically fail: softmax averaging of per-step probability distributions yields 1.835% gap (worse than either model only), and logit-space geometric averaging reaches 2.84%. This confirms that tour-level ensemble—where each model independently generates a complete tour and we select the shortest—is essential for sequence decisions.

InstanceNorm Calibration. We implement adaptive InstanceNorm calibration to mitigate distribution shift between original and augmented viewpoints. For each target normalization module, we collect source statistics from the original instance and target statistics from augmented viewpoints. The Wasserstein-2 distance between distributions is approximated via quantile matching over 100 bins. The blend coefficient for each channel is:

$$\lambda_c = \lambda_{min} + (1 - \bar{d}_c)^{\gamma} (\lambda_{max} - \lambda_{min})$$

where $\lambda_{min} = 0.05$, $\lambda_{max} = 0.50$, and $\gamma = 0.3$.

Adaptive Decision Modes. Under the strict constraint of exactly 8 augmentation viewpoints per instance (matching the baseline inference budget), we dynamically allocate viewpoints between the two models based on instance-specific feature distribution shifts. The standard deviation of Wasserstein shifts across layers (std) serves as the decision signal. We define three modes via grid-search thresholds (0.10, 0.14) on the validation set: P1 (std < 0.10): plain 4+4 split without norm; P2 (std >= 0.14): norm-enhanced 4+4 split; P3 (0.10 <= std < 0.14): plain 5+3 split favoring Model A. Each model independently decodes its assigned augmentation viewpoints, and the shortest tour across all 8 viewpoints is selected, achieving 0.478% gap—46.4% better than Model B alone and 76.7% better than the baseline.

4 Experiments

4.1 Experimental Setup

We evaluate on a held-out Test set of 10 TSPLIB instances (ch130, eil51, kroA150, kroB100, kroB200, kroD100, pr136, pr144, pr152, pr264). The performance of the proposed model is evaluated with optimality gap: gap = (score − optimal) / optimal × 100%. All results use 8-fold coordinate augmentation with argmax greedy decoding. No 2-opt, LKH, or random sampling is used. Each instance is solved exactly 8 times, matching the baseline inference budget.

4.2 Main Results

Table 1 reports results on the held-out Test set. The original POMO baseline achieves 2.048% average gap. Model A only (preference post-training) reduces this to 1.245% (39.2% relative improvement). Model B only (off-policy training with geometry-aware data) achieves 0.891%, a 56.5% relative improvement. Our adaptive dual-model ensemble further reduces the gap to 0.478%, a 76.7% improvement over the baseline and 46.4% over Model B only, while maintaining the same 8-viewpoint inference budget.

Table 1: Main Results on Test Set (10 instances)

Method	Avg Gap	Rel. Imp.	Win/Tie/Loss
Original POMO	2.048%	-	-
Model A	1.245%	+39.2%	7/0/3
Model B	0.891%	+56.5%	6/4/0
Adaptive Ensemble	**0.478%**	**+76.7%**	**6/4/0**

Rel. Imp. is relative improvement over Original POMO baseline.
Win/Tie/Loss counts vs baseline.

4.3 Ablation Studies

Table 2 presents ablation results isolating the contribution of each component on the Test set. All ensemble variants use exactly 8 augmentation viewpoints (the same inference budget), ensuring fair comparison.

Table 2: Ensemble Ablation on Test Set

Configuration	Avg Gap	Delta	Imp./Deg.
Original POMO	2.048%	-	-
Model A only	1.245%	-0.803	7/0.3
Model B only	0.891%	-1.157	6/4/0
Softmax step-wise average (A+B)	1.835%	-0.213	3/0/7
Fixed 4+4 split	0.545%	-1.503	5/3/2
Adaptive norm + fixed 4+4	0.535%	-1.513	6/2/2
Calibration-Guided Adaptive (Full)	**0.478%**	**-1.570**	**6/4/0**

Delta is change in percentage points from Original POMO baseline.
Imp./Deg. counts improved/degraded instances vs baseline.

4.4 Component Analysis

Model A Contribution. The preference post-training with curriculum learning reduces the gap from 2.048% to 1.245% (39.2% relative improvement), with 7 out of 10 test instances improving over the original POMO. The frozen reference policy ensures stable optimization, and the RL stabilization term prevents collapse to degenerate solutions. The LLM4AD-guided portfolio search further refines the inference strategy to plain argmax decoding with layer4 normalization calibration.

Model B Contribution. The off-policy training pipeline, guided by LLM4AD-evaluated data engineering strategies, achieves 0.891% gap, a 56.5% improvement over the baseline. The geometry-aware data generation specifically targets four dominant failure modes identified through TSPLIB diagnosis, with iterative recipe tuning from pilot to v2 informed by intermediate checkpoint evaluation. The compact5 inference strategy, discovered through LLM4AD-driven search, further enhances Model B by combining plain argmax with four norm-calibrated forward passes.

Ensemble Contribution. The calibration-guided adaptive ensemble achieves 0.478% gap, extracting complementary strengths from two independently trained models. The key insight is that InstanceNorm displacement statistics encode the impact of augmentation transformations on internal representations, allowing us to predict which model will perform better on each instance before decoding. Under the fixed 8-viewpoint budget, combining Model A (strong on medium-scale) and Model B (strong on large-scale) yields a synergistic effect: the ensemble outperforms both models individually, with a 6/4/0 improvement/tie/loss record on the test set. This demonstrates that tour-level ensemble of diverse policies is essential for sequence decisions, as step-level fusion catastrophically fails (1.835% gap).

5 Conclusion

We present a dual-model ensemble approach for neural TSP solvers that combines two independently trained POMO models with complementary strengths. Model A is refined via preference-based post-training with curriculum learning, with its inference strategy optimized by LLM4AD-driven portfolio search. Model B is trained through an off-policy pipeline with LLM4AD-guided data engineering and geometry-aware data synthesis. Our adaptive InstanceNorm calibration mechanism dynamically assigns augmentation viewpoints between the two models under a fixed 8-viewpoint budget. On a held-out test set of 10 TSPLIB instances, the ensemble achieves an average optimality gap of 0.478%, a 76.7% relative improvement over the original POMO baseline. Ablation studies demonstrate that each component contributes meaningfully and that tour-level ensemble of diverse policies is essential for sequence decisions.

Future directions include developing more realistic grid-like instance generators, extending LLM4AD-guided search to larger model ensembles with more than two policies, and applying the calibration framework to other NCO problems such as CVRP and OP.

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